

PROJECT REPORT

Physics-Constrained Telemetry Processing and Data Imputation Framework

A Robust Data Processing and Feature Engineering Pipeline for Multi-Stage Component Reliability Screening

Core Function	Telemetry cleaning, imputation, feature engineering and downstream ML integration
Testing Milestones	0h, 24h, 96h and 168h
Imputation Method	SAITS + Arrhenius-based physics constraints
Feature Engineering	MAD ratios, Rate-of-Change and multi-sensor covariance
Downstream Models	TCN / Attention-LSTM, Isolation Forest and GNN

1. Abstract

Modern hardware reliability screening depends on telemetry collected at multiple testing milestones. However, sensor dropouts and incomplete measurements can introduce gaps that disrupt temporal modelling and reduce the reliability of downstream anomaly-detection systems. This project proposes a physics-constrained telemetry processing framework that restores missing measurements while preserving both temporal continuity and physical plausibility.

The framework combines Self-Attention-Based Imputation for Time Series (SAITS) with Arrhenius-based thermal aging constraints. After imputation, the system generates dynamic features including lot-specific Median Absolute Deviation (MAD) ratios, instantaneous Rate-of-Change (RoC), and multi-sensor covariance features. The resulting complete feature set is supplied to TCN/Attention-LSTM forecasting, Isolation Forest anomaly detection, and Graph Neural Network (GNN) based analysis.

The overall objective is to create a robust preprocessing layer capable of maintaining uninterrupted data flow for component health assessment, degradation forecasting, batch anomaly detection, and physics-informed regression.

2. Problem Statement

Multi-stage hardware testing produces telemetry at different milestones such as 0h, 24h, 96h and 168h. If a sensor or acquisition system fails at an intermediate stage, the resulting missing data can break the expected sequence and negatively affect time-series, anomaly-detection and graph-based models.

A simple statistical interpolation may produce a numerically valid value but can also generate a degradation trajectory that is inconsistent with known thermal-aging behaviour. Therefore, a reliable solution must address both data completeness and physical consistency.

3. Objectives

- Detect and handle missing telemetry in multi-stage component testing.
- Restore missing intermediate measurements without breaking temporal sequence integrity.
- Combine attention-based time-series imputation with physics-based degradation constraints.
- Engineer features that improve the separability of normal and anomalous components.
- Capture lot-level, temporal and multi-sensor relationships.
- Provide complete inputs to forecasting, anomaly-detection and GNN modules.
- Support reliable hardware screening and early identification of degradation patterns.

4. System Architecture

The system is organized as a sequential data-processing pipeline. Raw telemetry first passes through cleaning and normalization. Missing values are detected and reconstructed by the physics-constrained imputation module. The completed sequence is then transformed into dynamic features before being distributed to the downstream machine-learning modules.

Stage	Processing	Output
1	Raw multi-stage telemetry	0h / 24h / 96h / 168h measurements
2	Cleaning and normalization	Validated and normalized data
3	Missing-data detection	Identified telemetry gaps
4	SAITS + Arrhenius constraint	Physically consistent complete sequence

Stage	Processing	Output
5	Dynamic feature engineering	MAD, RoC and covariance features
6	Module integration	Inputs for forecasting, anomaly detection and GNN
7	Final assessment	Health, drift and anomaly indicators

Overall data flow: Raw Telemetry → Cleaning → Normalization → Missing Data Detection → Physics-Constrained Imputation → Feature Engineering → ML Models → Component Health / Anomaly Assessment

5. Methodology

5.1 Data Collection and Preprocessing

Telemetry is collected at predefined testing milestones of 0h, 24h, 96h and 168h. The preprocessing stage removes invalid observations, standardizes measurement formats and normalizes relevant numerical variables. The system then identifies missing values and records their temporal position before model processing.

5.2 Physics-Constrained Imputation

The imputation module is designed to reconstruct a missing intermediate observation, such as the 96h telemetry, using the available observations at 0h, 24h and 168h. SAITS provides attention-based learning of temporal and cross-parameter relationships.

5.3 Bi-Directional Attention

Self-attention enables the model to use information from both earlier and later observations. This is useful for an intermediate missing point because the model can learn the component's behaviour before and after the missing milestone rather than depending only on the previous measurement.

- Learns relationships between multiple telemetry parameters.
- Uses information from both temporal directions.
- Captures non-linear and long-range dependencies.
- Produces a data-driven estimate for the missing measurement.

5.4 Arrhenius-Based Physical Constraint

The data-driven estimate is bounded using thermal-aging behaviour. The Arrhenius relationship can be expressed as:

$$k = A \exp(-E_a / (k_B T))$$

where k is the degradation or reaction rate, A is the pre-exponential factor, E_a is activation energy, k_B is the Boltzmann constant and T is absolute temperature. For accelerated burn-in testing around 125°C, the physical constraint prevents the reconstructed telemetry from producing unrealistic degradation behaviour.

Important implementation note: temperature must be converted to Kelvin when used in the Arrhenius equation. The physics constraint should be implemented as a model-specific bound or loss/optimization constraint using the appropriate parameter units and experimentally established aging limits.

6. Dynamic Feature Engineering

6.1 Lot-Specific MAD Ratios

Median Absolute Deviation provides a robust measure of dispersion around the lot median. For observations x_i , MAD is calculated as:

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$$

A component can then be compared against the behaviour of its exact manufacturing lot. This makes the anomaly measure more context-aware than comparing every component against a single global threshold.

6.2 Instantaneous Rate-of-Change

The rate at which a telemetry parameter changes between adjacent testing windows is represented as:

$$\text{RoC} = \Delta V / \Delta t = (V_{t_2} - V_{t_1}) / (t_2 - t_1)$$

RoC features can reveal abrupt changes even when the absolute value of a parameter remains within a nominal range. They are therefore useful for detecting sudden leakage-current increases, propagation-delay changes and other early-stage degradation patterns.

6.3 Multi-Sensor Covariance

Covariance analysis captures dependencies between multiple telemetry parameters. Instead of analysing each sensor independently, the framework examines whether several parameters change together. For example, a simultaneous increase in temperature, leakage current and propagation delay may indicate a developing component anomaly.

- Detects multivariate anomalies.
- Captures hidden dependencies between sensors.
- Improves anomaly separability.
- Complements univariate threshold-based checks.

7. Integration with Downstream Modules

7.1 Solution 1: Temporal Forecasting and Ensemble Anomaly Detection

The cleaned telemetry sequence and engineered RoC vectors are supplied to a TCN/Attention-LSTM drift forecaster and an Isolation Forest ensemble.

- **TCN / Attention-LSTM:** learns temporal degradation patterns and predicts future drift or health trajectories.
- **Isolation Forest:** detects observations or components that differ significantly from expected feature distributions.

7.2 Solution 2: GNN-Based Spatial and Batch Analysis

The completed node matrices are used to update the GNN representation. Since missing telemetry has already been reconstructed, the graph does not contain an artificial gap at the 96h stage.

- Supports graph-based batch anomaly detection.
- Preserves relationships between component nodes.
- Supports physics-informed regression.
- Allows downstream graph analysis to continue without missing-node interruption.

8. Algorithmic Workflow

1. Acquire telemetry from all available testing milestones.

2. Clean invalid measurements and normalize numerical parameters.
3. Identify missing telemetry and its corresponding timestamp.
4. Use SAITS to estimate the missing measurement from temporal and cross-parameter information.
5. Apply Arrhenius-based physical bounds to the estimated value.
6. Construct the complete telemetry sequence.
7. Calculate lot-specific MAD ratios.
8. Calculate instantaneous Rate-of-Change features.
9. Construct multi-sensor covariance features.
10. Feed temporal features to TCN/Attention-LSTM and Isolation Forest.
11. Update GNN node matrices with the complete telemetry.
12. Generate final degradation, anomaly and component-health indicators.

9. Expected Results

- Continuous telemetry sequences despite intermediate sensor dropouts.
- Physically plausible reconstruction of missing measurements.
- Improved detection of abrupt and multivariate anomalies.
- Better identification of lot-specific outliers.
- Improved temporal degradation forecasting.
- Uninterrupted GNN-based component and batch analysis.
- More robust hardware screening and reliability assessment.

10. Advantages

Advantage	Description
Physics-aware	Combines machine-learning imputation with thermal-aging constraints.
Temporal continuity	Maintains the expected sequence across testing milestones.
Robust statistics	Uses MAD to reduce sensitivity to extreme outliers.
Early anomaly detection	RoC features expose sudden changes in behaviour.
Multivariate analysis	Covariance captures cross-sensor dependencies.
Model compatibility	Provides complete inputs for temporal, tree-based and graph models.
Scalable architecture	The preprocessing layer can serve multiple downstream analytical modules.

11. Implementation Considerations

- Arrhenius parameters such as activation energy should be selected from appropriate component/material reliability data rather than assumed universally.
- Physics bounds should be calibrated using experimental burn-in data and the expected degradation mechanism.
- Imputation quality should be validated by masking known telemetry values and comparing reconstructed values with ground truth.
- The framework should retain an explicit indicator showing which values are measured and which are imputed.

- Uncertainty estimates should be considered because an imputed value is not equivalent to a directly measured observation.
- Sensor calibration, unit consistency and timestamp synchronization are essential before feature engineering.

12. Conclusion

The proposed Physics-Constrained Telemetry Processing and Data Imputation Framework provides a robust foundation for handling incomplete multi-stage hardware telemetry. By combining SAITS-based attention imputation with Arrhenius-based thermal-aging constraints, the system can reconstruct missing intermediate measurements while maintaining physically meaningful degradation behaviour.

The subsequent feature-engineering stage strengthens anomaly detection through lot-specific MAD ratios, instantaneous Rate-of-Change and multi-sensor covariance. These features can be integrated with temporal forecasting, ensemble anomaly detection and graph-based analysis.

Overall, the framework creates a continuous and physics-aware data pipeline that can improve component screening, degradation monitoring and early anomaly identification in reliability-focused hardware testing environments.

13. End-to-End Summary

Layer	Technology / Method	Primary Purpose
Input	Multi-stage telemetry	Collect component measurements
Preprocessing	Cleaning + normalization	Prepare reliable inputs
Imputation	SAITS	Recover missing temporal data
Constraint	Arrhenius kinetics	Maintain physical plausibility
Features	MAD + RoC + covariance	Increase anomaly separability
Solution 1	TCN / Attention-LSTM + Isolation Forest	Forecast drift and detect anomalies
Solution 2	GNN + physics regression	Analyse component relationships and batch anomalies
Output	Health / anomaly indicators	Support reliability screening